

AJAY KUMAR MEENA

Gwalior, Madhya Pradesh

+91-9319636494 | Personal:aj007026180@gmail.com | College:imt_2022010@iiitm.ac.in | LinkedIn | GitHub | LeetCode | Codeforces

Education

Indian Institute of Information Technology and Management Gwalior

2022 – 2027

Integrated B.Tech + M.Tech in Information Technology

Madhya Pradesh, India

CGPA: 8.41/10.0

Experience

Software Engineering Intern

May 2026 – July 2026

Flipkart

Bengaluru, India

- Engineered an autonomous AI software agent workflow integrating LLMs (Claude API) with live codebase context gathered via Model Context Protocol (MCP) to automatically resolve JIRA tickets into GitHub PRs.
- Built backend orchestration pipelines using Celery and Redis to manage multi-agent tasks, enabling automated Java code generation and isolated Gradle test execution within Docker containers.
- Designed a cross-process human-in-the-loop ML architecture featuring a real-time web dashboard (FastAPI, React) allowing engineers to securely evaluate, edit, and approve AI-generated reasoning plans mid-flight.
- Deployed a highly available Spring Boot service utilizing a DAG-based workflow for Flipkart's Returns platform, orchestrating step sequencing and failure handling across the returns processing flow.

Research Intern

May 2025 – April 2026

IIT Mandi iHub and HCl Foundation

Mandi, India

- Developed deep learning models for Human Activity Recognition (HAR), addressing domain shift challenges through feature extraction from raw multidimensional time-series accelerometer and gyroscope data.
- Architected and trained Temporal Convolutional Networks (TCNs), conducting rigorous empirical evaluations and ablation studies against both public benchmarks and internally curated datasets.
- Implemented on-device ML workflows by optimizing and quantizing models into TensorFlow Lite, integrating them into Android applications for low-latency, real-time edge inference.
- Researched and structured pipelines compatible with Federated Learning to enable privacy-preserving, decentralized model fine-tuning on resource-constrained edge devices.

Key Projects

Resonance: Multi-Modal Two-Stage Recommendation Engine

 **Live**

PyTorch, Transformers, FAISS, EASE, Data Aggregation, InfoNCE Loss

- Engineered a recommendation system utilizing a Tri-Modal Two-Tower Neural Network with self-attention and GRUs to fuse rich textual, Knowledge Graph, and statistical embeddings.
- Designed an extreme-scale retrieval layer applying Collaborative Filtering (EASE) and geometric vector search (FAISS), optimizing contrastive InfoNCE loss to achieve a Retrieval Recall of **57.46%** across 200K+ users.
- Deployed a Stage-2 Neural Reranker using Sequence-wise Multi-Head Self-Attention to enforce diverse slate contextualization, securing an End-to-End Approximate Hit Rate (HR@10) of **21.5%**.
- Processed the MovieLens 32M dataset and integrated custom TMDb/IMDb metadata pipelines to generate 9,600+ enriched movie profiles via automated thematic summarization.

Momentum: Privacy-Preserving Federated Learning System

 **GitHub**

Python, TensorFlow, TFLite (Flex Delegate), Distributed ML, Kotlin

- Engineered an end-to-end Federated Learning framework natively supporting on-device training to solve personalization and domain-adaptation bottlenecks in smartphone sensor-based activity recognition.
- Trained a baseline Temporal Convolutional Network to **99.6%** benchmark accuracy and mapped the computational graph into a trainable TensorFlow Lite artifact for local client execution.
- Developed the secure client node infrastructure via a native Kotlin application, facilitating real-time inference and user-driven active learning to supply high-quality ground truth for decentralized gradient updates.

Technical Skills

Machine Learning & AI

PyTorch, TensorFlow, Transformers, RecSys, Federated Learning, LangGraph

Data Science & Search

Pandas, NumPy, Scikit-learn, FAISS, RAG, MCP, Vector Embeddings

Programming Languages

Python, C/C++, Java, Go, TypeScript, SQL

Systems & Infrastructure

Docker, AWS, FastAPI, Spring Boot, PostgreSQL, Redis

Achievements

- Secured **1st rank** in college and **5th overall** in *CodeRush*, IIITM Gwalior's coding competition. [\[team_404\]](#)
- Solved 1000+ DSA problems; achieved peak competitive programming ratings of **1865** on LeetCode and **1298** on Codeforces.
- Reached the Semi-Finals of the Flipkart GRID Hackathon, placing in the Top **1%** of over 160,000 national participants.